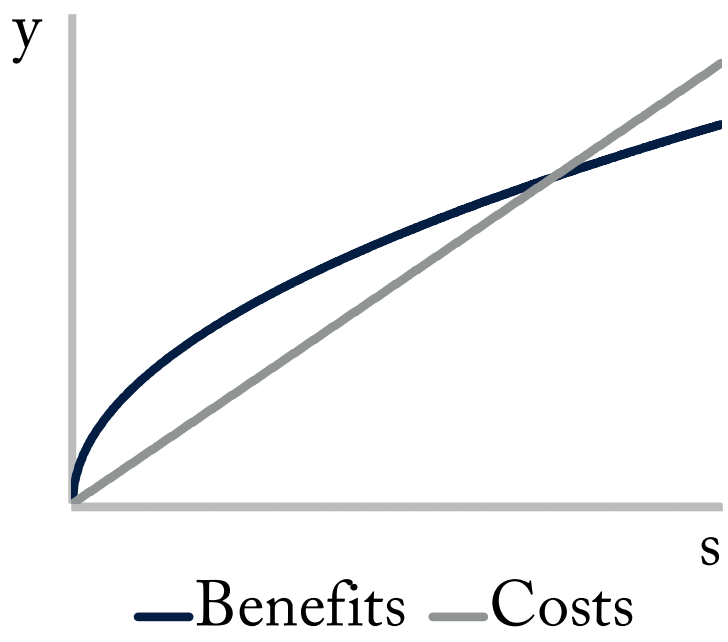
GEORGETOWN UNIVERSITY

A Mincerian Argument for Vocational Education

Richard Newsome

Modified Mincer Equation



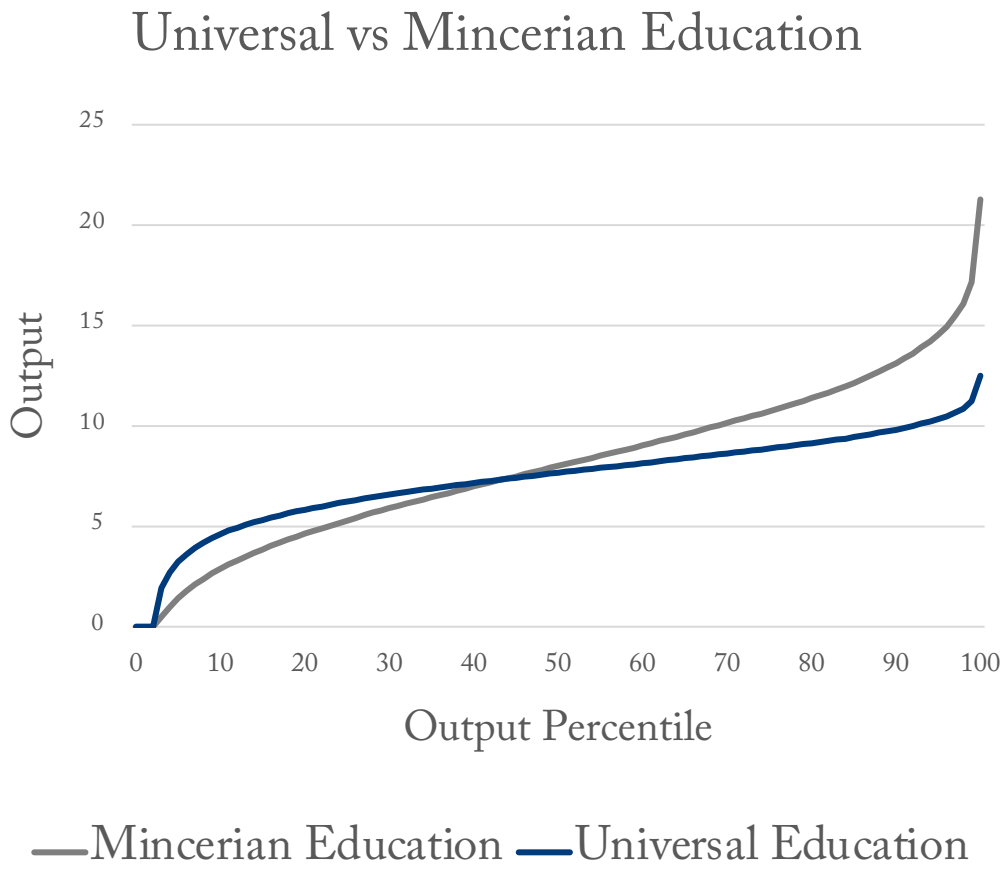
$$y = \sqrt{s\theta} - sc$$

$$s_{optimal} = \frac{\theta}{4c^2}$$

$$y_{optimal} = \sqrt{s_{optimal}\theta} - s_{optimal}c$$

$$y_{optimal} = \frac{\theta}{4c}$$

Compared to Universal Education



Model parameters:

$c=1$

θ: mean=32, sd=16

Mincerian Education:

- $y_{avg} = 8.04$
- $Gini_y = 0.276$

Universal Education:

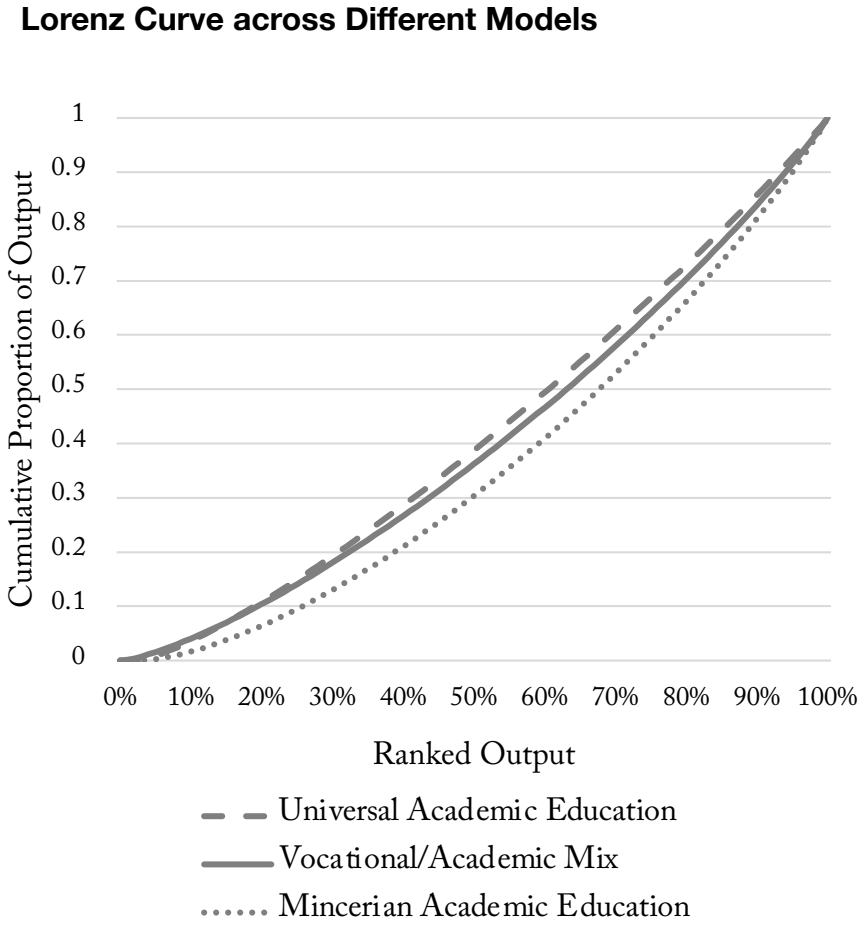
- $y_{avg} = 7.35$
- $Gini_y = 0.163$

Allowing for Vocational Education

Academic Education Only	Academic + Vocational Education
$y = \sqrt{s\theta} - sc$	$y = \sqrt{\frac{1}{2}s\theta} + \sqrt{\frac{1}{2}v\phi} - (s+v)c$
$s_{optimal} = \frac{\theta}{4c^2}$	$s_{optimal} = \frac{\theta}{8c^2}$ $\quad v_{optimal} = \frac{\phi}{8c^2}$
$y_{optimal} = \frac{\theta}{4c}$	$y_{optimal} = \frac{\theta + \phi}{8c}$

Inequality and Output across Three Models

	Gini Coefficient	Average Output
Mincerian Academic Education	0.276	8.04
Universal Academic Education	0.163	7.35
Vocational/Academic Mix	0.195	8.04



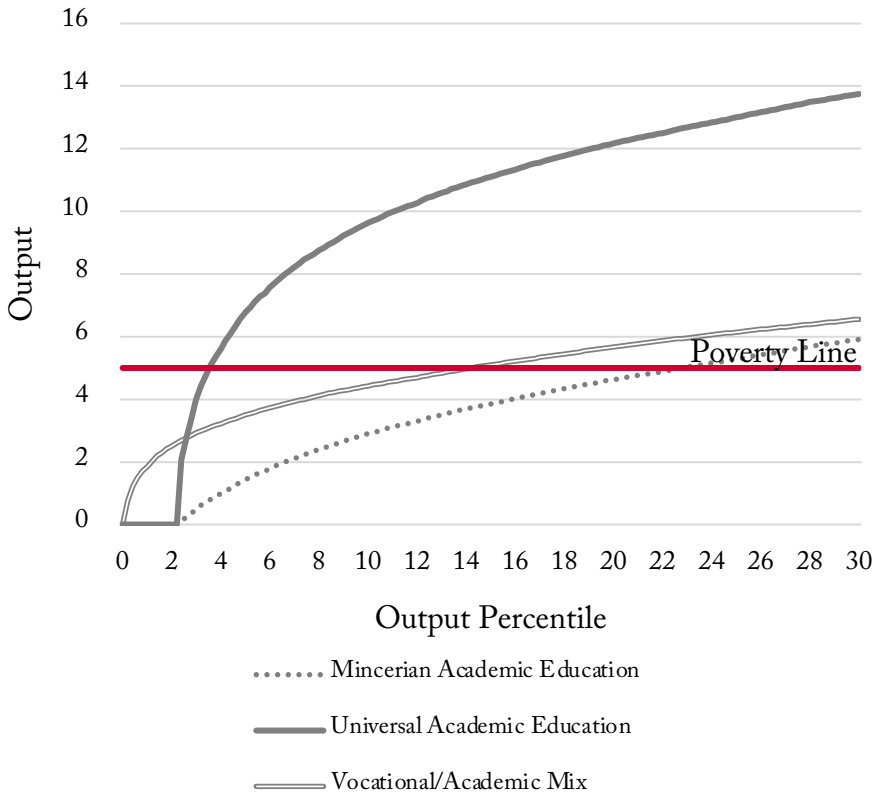
Poverty Across Models

FGT Poverty Statistics under z=5

	Mincerian Academic	Universal Academic*	Vocational/ Academic
FGT_0	22.8%	3.5%	14.2%
FGT_1	9.8%	2.6%	3.7%
FGT_2	6.5%	2.4%	1.6%

*Before cost covering

Poverty Profiles – Left-hand Portion



Concluding Remarks

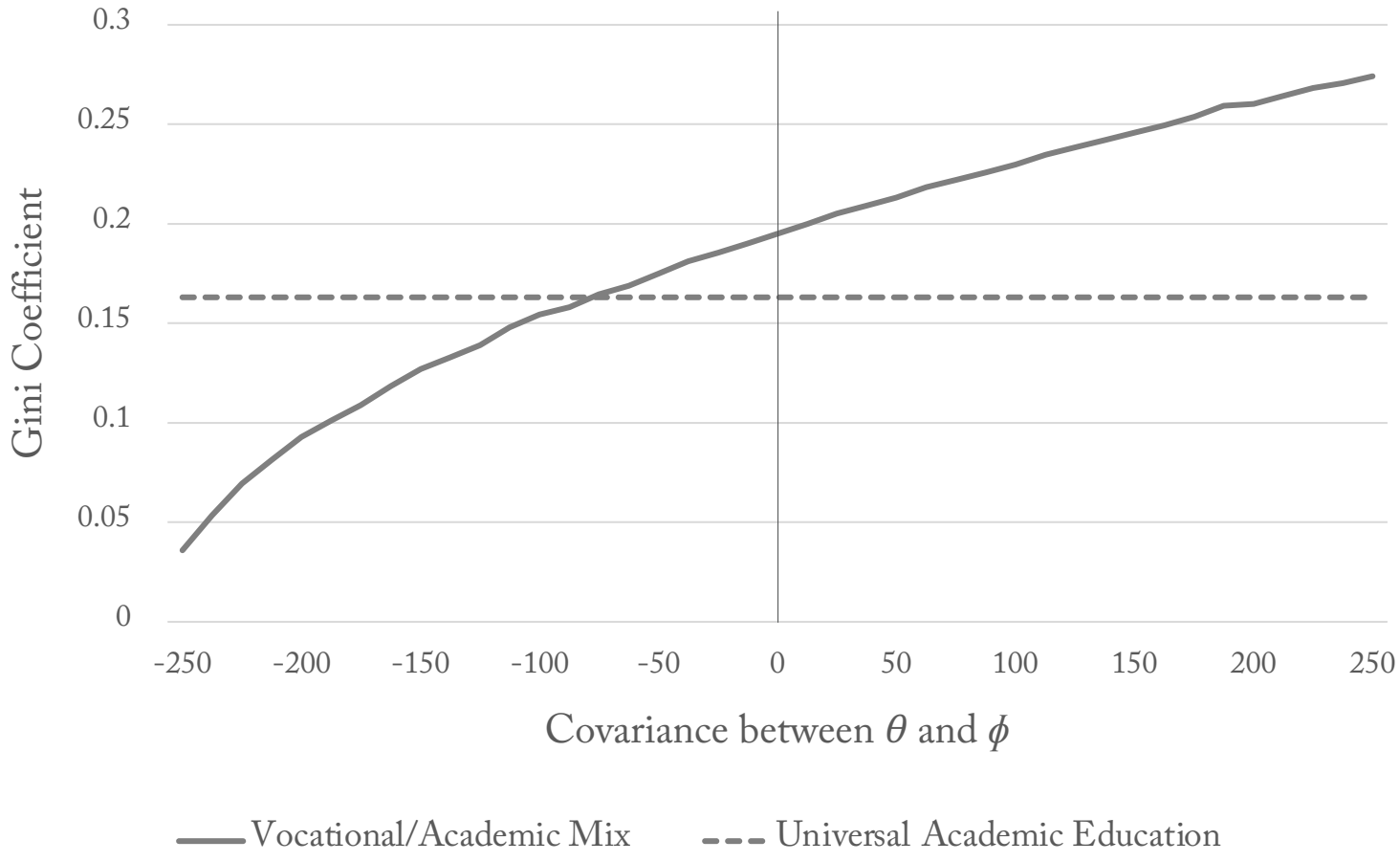
- Vocational/Academic Education
  - Reduces economic inequality
  - Reduces poverty, especially extreme poverty
  - Faces no penalty to total output
- Universal Academic Education
  - Reduces economic inequality
  - Reduces poverty, but less so extreme poverty
  - Does so at the expense of lowering average output by 8.6%
- All of this is places significant assumptions on our functional form

$$y = \sqrt{\frac{1}{2}s\theta} + \sqrt{\frac{1}{2}v\phi} - (s+v)c$$

$$y_{optimal} = \frac{\theta + \phi}{8c}$$

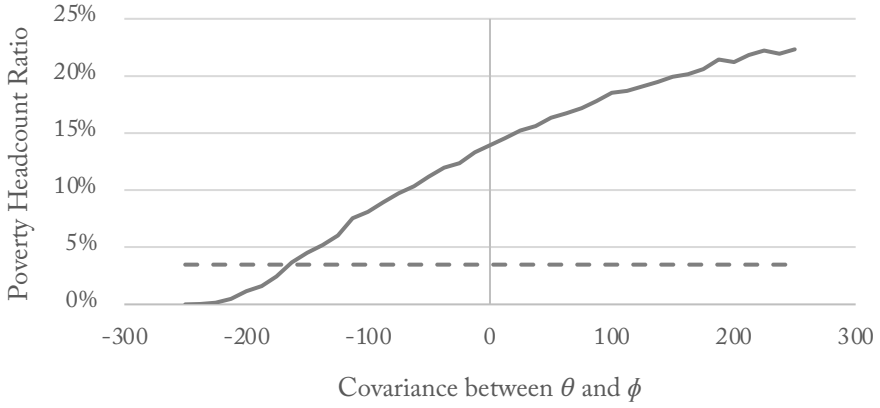
Challenging Assumptions

Covariance in Ability - Inequality

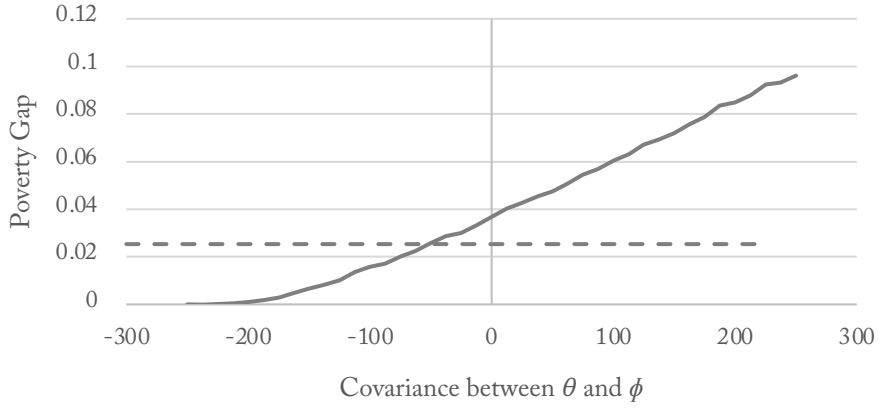


Covariance in Ability - Poverty

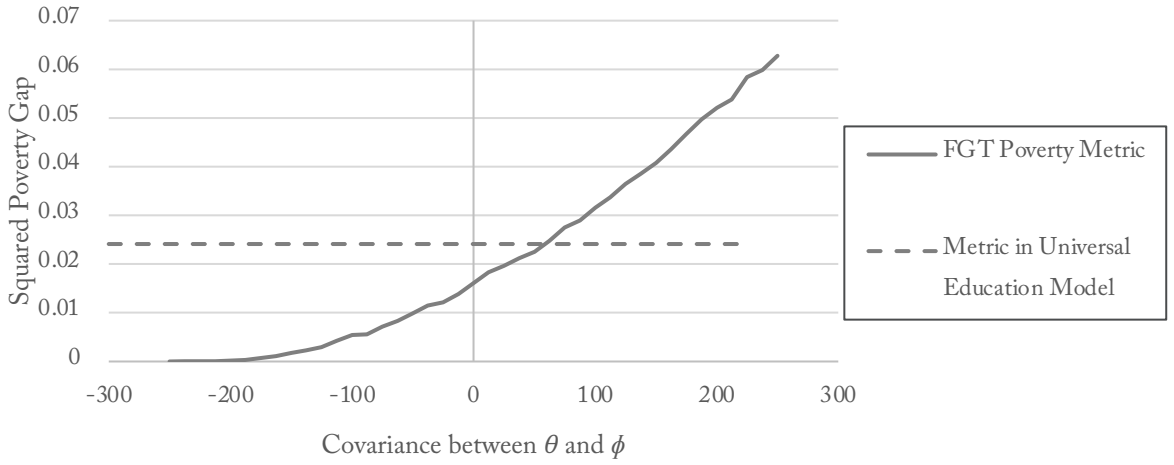
α=0



α=1



α=2



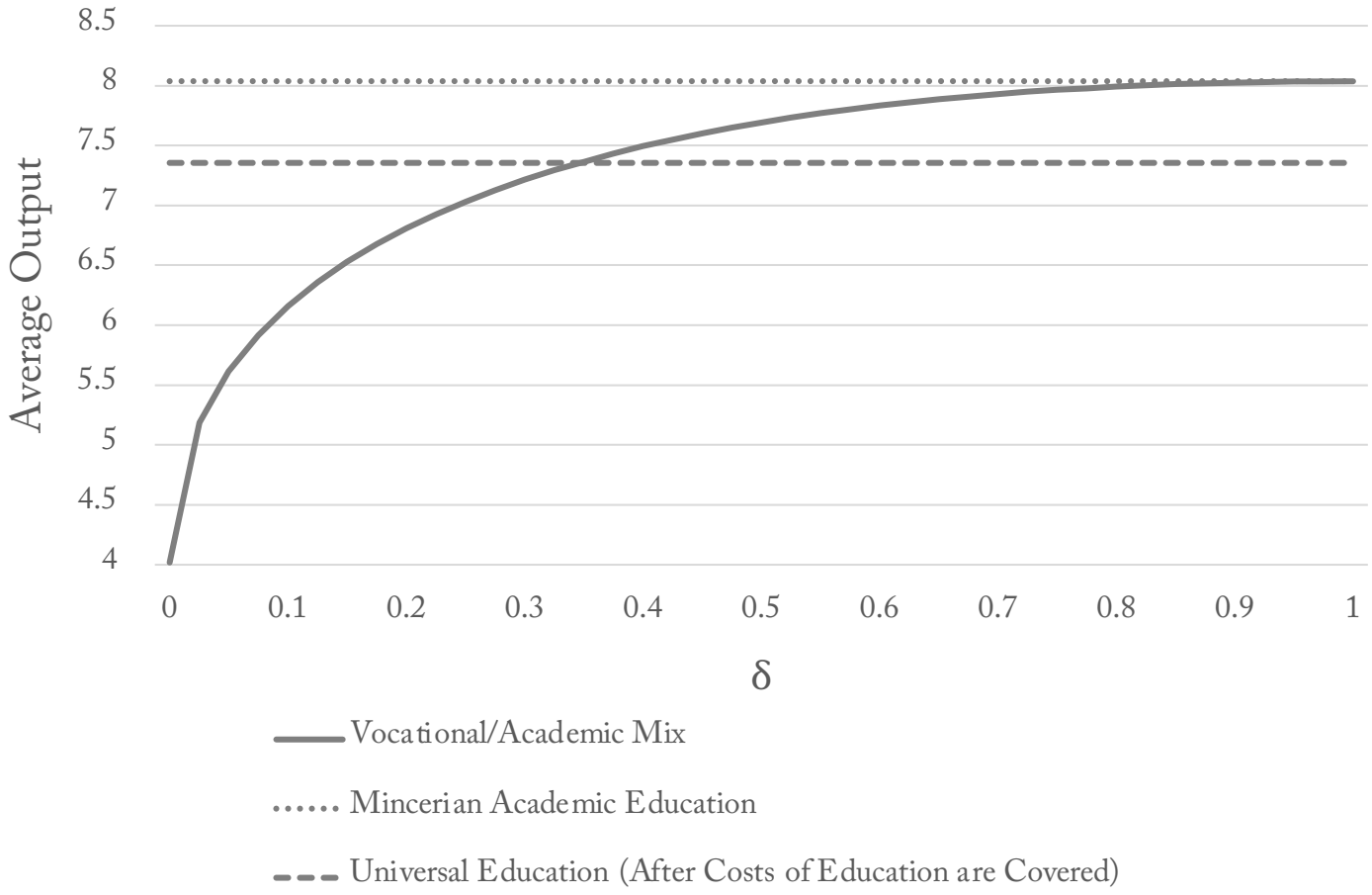
Undervaluation of Vocational Ed.

$$v = \delta v_{optimal} = \delta \frac{\phi}{8c^2}$$

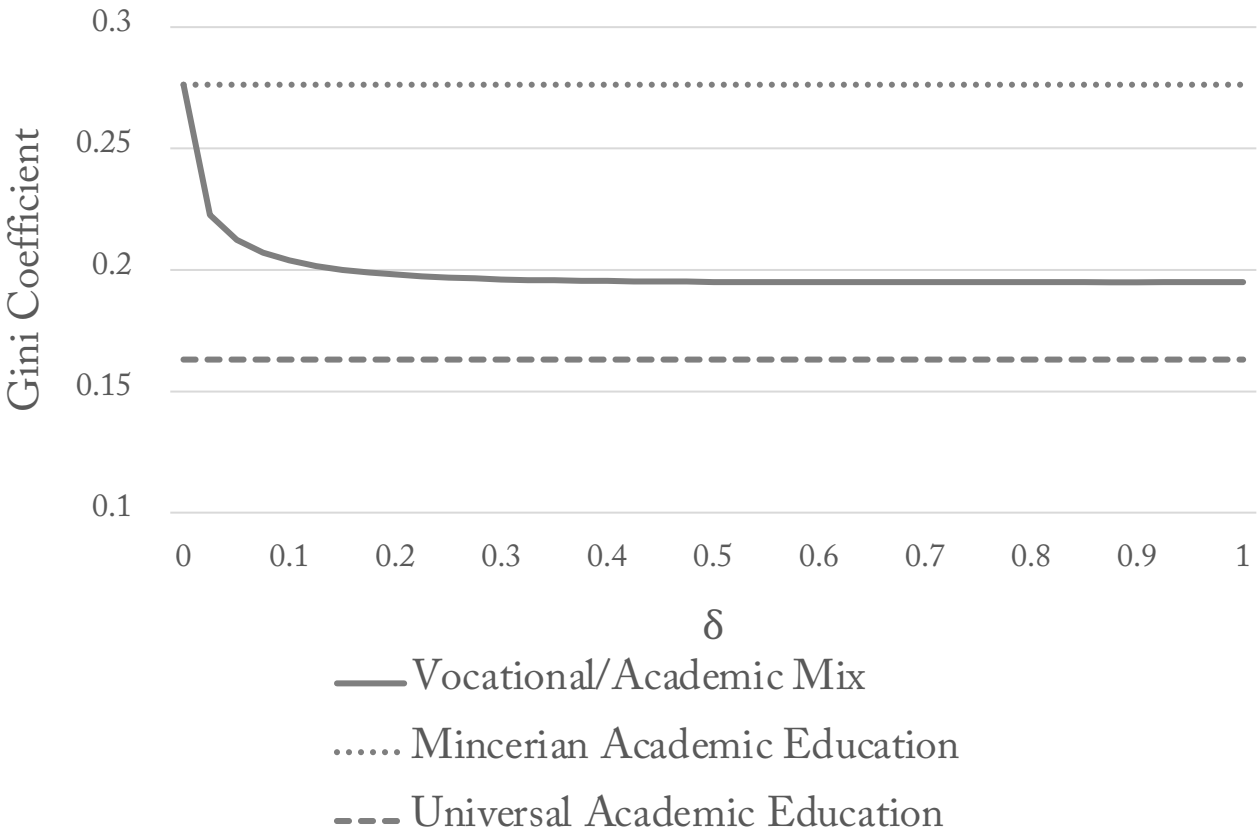
$$y = \sqrt{\frac{1}{2} s_{optimal} \theta} + \sqrt{\frac{1}{2} \delta v_{optimal} \phi} - \left(s_{optimal} + \delta v_{optimal}\right) c$$

$$y = \frac{\theta}{8c} + \frac{\phi}{8c}\left(2\sqrt{\delta} - \delta\right)$$

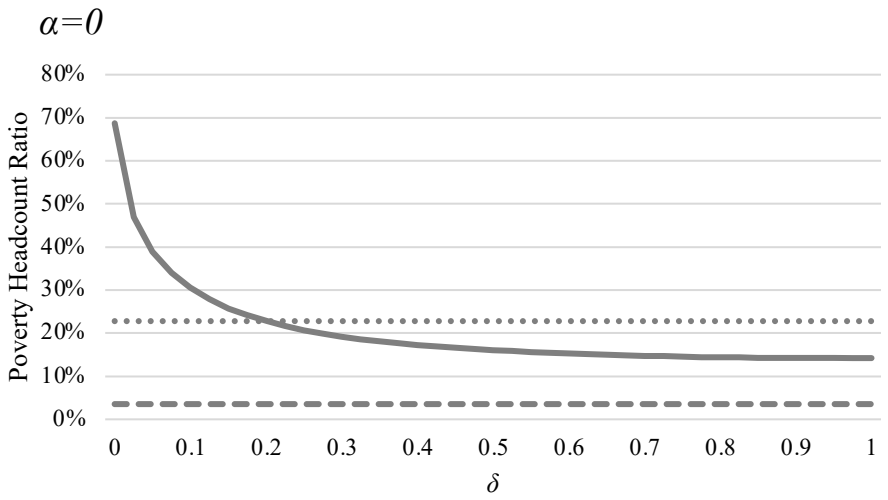
Undervaluation of Vocational Ed. - Output

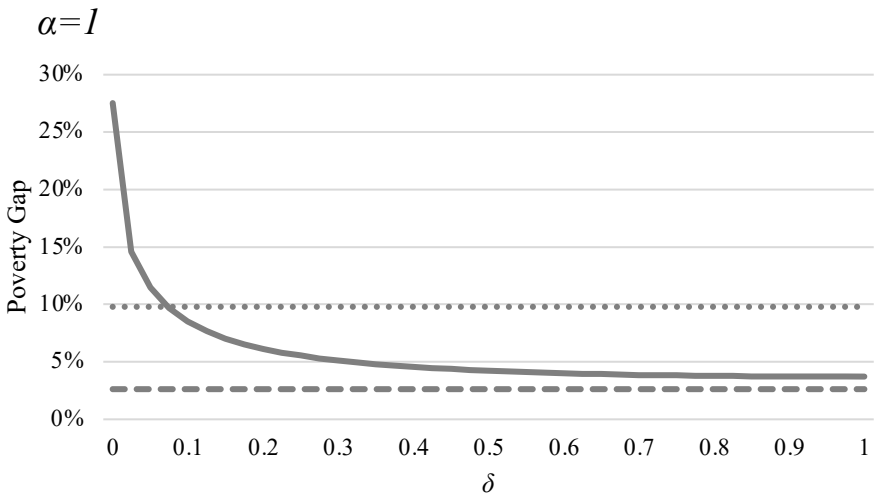


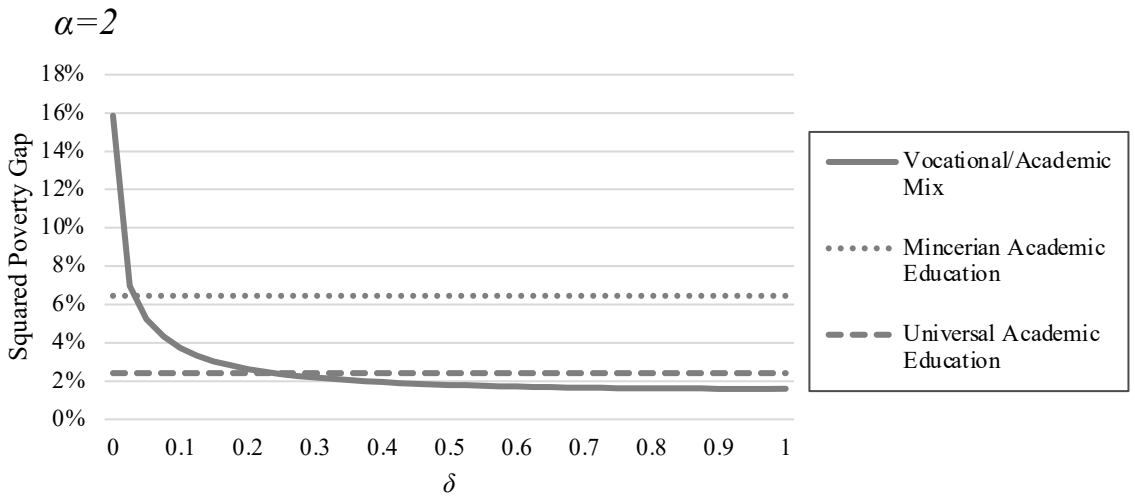
Undervaluation of Vocational Ed. - Inequality



Undervaluation of Vocational Ed. - Poverty







Thank you!

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